

ESP32 Star Network - Sensors and Actuators GPIO Mapping

Node	Type	Component	GPIO Pin
Zone 1A	Sensor	DHT11 Temperature & Humidity	GPIO 4
Zone 1A	Sensor	Soil Moisture Sensor	GPIO 34
Zone 1A	Actuator	Water Pump	GPIO 23
Zone 1A	Actuator	Red LED	GPIO 25
Zone 1A	Actuator	Green LED	GPIO 26
Zone 1A	Actuator	Blue LED	GPIO 27
Zone 1B	Sensor	Touch Sensor	GPIO 4
Zone 1B	Sensor	PIR Motion Sensor	GPIO 18
Zone 1B	Sensor	Rain Sensor	GPIO 34
Zone 1B	Actuator	Roof Protection Servo	GPIO 23
Zone 2A	Sensor	DHT11 Temperature & Humidity	GPIO 4
Zone 2A	Sensor	RFID SDA(SS)	GPIO 21
Zone 2A	Sensor	RFID RST	GPIO 27
Zone 2A	Sensor	RFID SCK	GPIO 18
Zone 2A	Sensor	RFID MISO	GPIO 19
Zone 2A	Sensor	RFID MOSI	GPIO 23
Zone 2A	Sensor	RTC IO	GPIO 25
Zone 2A	Sensor	RTC CLK	GPIO 26
Zone 2A	Sensor	RTC RST	GPIO 14
Zone 2A	Actuator	Buzzer	GPIO 32
Zone 2A	Actuator	LED 1	GPIO 12
Zone 2A	Actuator	LED 2	GPIO 2
Zone 2A	Display	OLED SDA	GPIO 13
Zone 2A	Display	OLED SCL	GPIO 22
Zone 2B	Sensor	Ultrasonic Trigger	GPIO 5
Zone 2B	Sensor	Ultrasonic Echo	GPIO 18
Zone 2B	Sensor	LDR Sensor	GPIO 34
Zone 2B	Sensor	HX711 DT	GPIO 32
Zone 2B	Sensor	HX711 SCK	GPIO 4
Zone 2B	Actuator	Ventilation Fan	GPIO 27
Zone 2B	Actuator	Lighting LED	GPIO 23
Zone 2B	Actuator	Feed Dispensing Servo	GPIO 19
Zone 3A	Sensor	Water Level LDR 1	GPIO 35

Zone 3A	Sensor	Water Level LDR 2	GPIO 34
Zone 3A	Sensor	TDS Sensor	GPIO 32
Zone 3A	Actuator	Water Valve Servo	GPIO 23
Zone 3A	Actuator	Water Pump	GPIO 15
Zone 3A	Actuator	Status LED	GPIO 2
Zone 3B	Sensor	DS18B20	GPIO 4
Zone 3B	Sensor	Optical Water Level Sensor	GPIO 34
Zone 3B	Actuator	Alert LED	GPIO 15
Zone 3B	Actuator	Buzzer	GPIO 2
Zone 3B	Display	SH1106 OLED	I ² C